

Numerical study of an energy cascade model derived from a dispersive equation

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Abstract

TODO

1 Introduction

The transfer of energy between the modes of a nonlinear Hamiltonian system is a basic question across the sciences, from fluids and oceanography to climate. When external forcing and dissipation act on a few modes while the bulk—the *inertial range*—evolves intrinsically, the natural framework is out-of-equilibrium statistical physics; for wave systems it is *wave* (or *weak*) *turbulence theory* [11], in which energy is transported between Fourier modes, that is, across spatial scales. The classical description proceeds through the wave kinetic equation, whose power-law stationary solutions are the Kolmogorov–Zakharov cascade spectra $n(k) = |k|^{-d}$ and $n(k) = |k|^{-d+2/3}$ (the direct and inverse cascades). These spectra are themselves nonequilibrium steady states (NESS), in which energy flows at constant flux through the inertial range; their rigorous justification remains largely open [2].

Following [2], we adopt a complementary viewpoint: rather than the kinetic equation, we study the invariant measure—the NESS itself—of a forced and dissipated reduction of the nonlinear Schrödinger (NLS) equation. The present paper is the numerical companion to that program. We compute and analyze the NESS of the energy cascade chain for the three-mode system treated analytically in [2] and for long chains, where the analysis is open.

1.1 Derivation of the model

Consider the cubic NLS on a torus,

$$-i\partial_t v + \Delta v = \pm |v|^2 v. \quad (1.1)$$

In Fourier variables a_K the cubic nonlinearity couples wavevectors with $K_1 - K_2 + K_3 = K$. Retaining only the *resonant* interactions, those with $|K_1|^2 - |K_2|^2 + |K_3|^2 = |K|^2$, gives the resonant NLS system, which approximates (1.1) for sufficiently small data. Colliander, Keel, Staffilani, Takaoka and Tao [1] restrict this system to a finite set of frequencies

$$\Lambda = \Lambda_1 \cup \Lambda_2 \cup \cdots \cup \Lambda_n \subset \mathbb{Z}^2,$$

on which the amplitude is constant within each generation Λ_j . The generations are produced by an explicit combinatorial “nuclear family” construction that embeds successive Λ_j in $\mathbb{Z}^2$ through exact rational (Pythagorean) rotations, with Λ_1 supported at low and Λ_n at high frequency.

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Writing $c_j(t)$ for the common amplitude on Λ_j , the resonant system collapses to the *toy model* of [1],

$$-i\dot{c}_j = 2\left(\sum_k |c_k|^2\right)c_j - |c_j|^2 c_j + 2(c_{j-1}^2 + c_{j+1}^2)\bar{c}_j, \quad j = 1, \dots, n, \quad c_0 = c_{n+1} = 0. \quad (1.2)$$

In [1] this model is shown to admit an energy-cascade solution whose mass, concentrated on Λ_1 at $t = 0$, migrates to Λ_n at a later time—a deterministic route to the growth of Sobolev norms of (1.1).

System (1.2) is Hamiltonian. In action–angle coordinates $c_j = \sqrt{I_j} e^{i\varphi_j}$, with the *bond angles* $\theta_k := 2(\varphi_k - \varphi_{k-1})$, it is generated by the Hamiltonian

$$H(\mathbf{I}, \boldsymbol{\varphi}) = \left(\sum_{k=1}^n I_k\right)^2 - \frac{1}{2} \sum_{k=1}^n I_k^2 + 2 \sum_{k=2}^n I_{k-1} I_k \cos \theta_k, \quad (1.3)$$

with the Hamiltonian equations of motion

$$-\partial_{\varphi_j} H = 4I_j(I_{j-1} \sin \theta_j - I_{j+1} \sin \theta_{j+1}), \quad \partial_{I_j} H = 2M - I_j + 2I_{j-1} \cos \theta_j + 2I_{j+1} \cos \theta_{j+1}, \quad (1.4)$$

where $M = \sum_k I_k$ and $I_0 = I_{n+1} = 0$.

Remark 1.1 (Normalization convention). The Hamiltonian (1.3) is twice the action–angle Hamiltonian of [2]; equivalently the drift (1.4) is twice theirs, and the equilibrium measure below carries the factor $1/2T$ rather than $1/T$. This amounts to a rescaling of time and is fixed throughout so that every formula matches the simulation code that produces our numerical data. For $n = 3$ with the middle mode as reference one writes $\theta_1 = 2(\varphi_1 - \varphi_2)$, $\theta_3 = 2(\varphi_3 - \varphi_2)$, recovering the reduced coordinates of [2].

To realize an energy cascade as a statistical steady state one injects energy at the low-frequency end and extracts it at the high-frequency end: the two end modes $j = 1$ and $j = n$ are coupled to stochastic heat baths at temperatures T_1 and T_n . This produces the Itô system

$$\begin{aligned} dI_j &= -\partial_{\varphi_j} H dt + b_j^I dt + \sigma_j^I dW_j^I, \\ d\varphi_j &= \partial_{I_j} H dt + b_j^\varphi dt + \sigma_j^\varphi dW_j^\varphi, \quad j = 1, \dots, n, \end{aligned} \quad (1.5)$$

in which the bath drift and diffusion (b, σ) are supported only on $j \in \{1, n\}$. We take the fluctuation–dissipation (Langevin) form: at the hot end $j = 1$,

$$b_1^I = 2\gamma(2T_1 - I_1 \partial_{I_1} H), \quad \sigma_1^I = 2\sqrt{2\gamma T_1 I_1}, \quad b_1^\varphi = -2\gamma I_2 \sin \theta_2, \quad \sigma_1^\varphi = \sqrt{2\gamma T_1 / I_1}, \quad (1.6)$$

and at the cold end

$$b_n^I = 2\gamma(2T_n - I_n \partial_{I_n} H), \quad b_n^\varphi = 2\gamma I_{n-1} \sin \theta_n,$$

with the corresponding replacements $T_1 \rightarrow T_n$ in the diffusion coefficients. Here $\gamma > 0$ is the coupling strength. The bath (1.6) is constructed so that at equilibrium $T_1 = T_n = T$ the system admits the explicit Gibbs invariant density

$$\rho_{\text{eq}}(\mathbf{I}, \boldsymbol{\varphi}) \propto \exp(-H(\mathbf{I}, \boldsymbol{\varphi})/2T). \quad (1.7)$$

Proposition 1.2 (Equilibrium consistency of the bath). *Assume $T_1 = T_n = T$ and impose the natural no-flux boundary condition at $I_j = 0$. For every smooth observable f that is periodic in the angles and compactly supported in the action variables,*

$$\int \mathcal{L}f(\mathbf{I}, \boldsymbol{\varphi}) \rho_{\text{eq}}(\mathbf{I}, \boldsymbol{\varphi}) d\mathbf{I} d\boldsymbol{\varphi} = 0,$$

where $\mathcal{L}$ is the generator of (1.5)–(1.6). Hence (1.7) is a stationary density for the equal-temperature process, provided the process is nonexplosive.

Proof. Split $\mathcal{L}$ into the Hamiltonian transport and the two endpoint bath operators. The Hamiltonian vector field is divergence-free in $(\mathbf{I}, \boldsymbol{\varphi})$ and conserves H , so its adjoint annihilates every density that is a function of H . At each endpoint, the drift and diffusion coefficients in (1.6) are in fluctuation–dissipation balance with inverse temperature $1/(2T)$; a direct integration-by-parts calculation, with the no-flux condition at $I_j = 0$, shows that the endpoint adjoint also annihilates $\exp(-H/2T)$. When $T_1 = T_n = T$, the two endpoint calculations use the same temperature, and summing the adjoint contributions gives $\mathcal{L}^* \rho_{\text{eq}} = 0$ in the weak sense above. $\square$

Remark 1.3 (Choice of thermostat). The analysis of [2] uses a different thermostat—dissipation of the form $\gamma(T_j - I_j^3)$ with a tailored multiplicative angle noise—chosen to make the Lyapunov construction tractable; under that choice the Gibbs measure (1.7) is *not* invariant, even at equal temperatures. For our numerical study we instead use the fluctuation–dissipation bath (1.6), whose equilibrium measure is the explicit Gibbs density (1.7); this provides a stringent check on the Fokker–Planck solver (section 4.1). The reduced-system analysis of the stabilization mechanism in section 4.2 retains the original $\gamma(T - I^3)$ bath of [2] for direct comparison with the proof.

1.2 Known results and open problems

For the three-mode chain ($n = 3$), in which only the middle mode I_2 is unforced, Hani, Li, Nahmod and Staffilani [2] establish the existence, uniqueness and ergodicity of the NESS: when the two bath temperatures are sufficiently different, the system possesses a unique invariant probability measure π , and the law of the process converges to π in total variation at a power-law rate in time. The proof rests on a novel construction of a Lyapunov function through an elliptic Feynman–Kac equation. The central difficulty is that, unlike microscopic heat conduction where energy simply flows from hot to cold, here the transfer of energy to and from the unforced middle mode depends sensitively on the phase variables being “correct”: the system must be prevented both from *overheating* (energy trapped in I_2) and from *freezing* (energy blocked at I_2), and which of these threatens depends on whether I_2 is large or small. This phase-dependent stabilization mechanism is the analytical heart of [2] and the object of our section 4.2.

Two limitations of the analytical theory motivate the present work. First, the results of [2] are confined to $n = 3$; the authors explicitly leave longer chains open, noting that already the three-mode case is demanding. Second, the NESS is not given in closed form, so quantitative properties—action profiles, transport coefficients, local statistics, the relaxation spectrum—are inaccessible to the existence theory. Both gaps are naturally addressed numerically.

1.3 Positioning within nonequilibrium transport

The present problem sits at the intersection of two related but distinct literatures. In wave turbulence, one studies nonequilibrium cascades generated by resonant wave interactions and their kinetic stationary spectra [11]. The cascade chain used here is instead a finite resonant subsystem of cubic NLS, inherited from the CKSTT construction [1] and from the three-mode NESS theory of [2]; the chain index is a generation label in frequency space rather than a physical lattice coordinate. In low-dimensional nonequilibrium statistical mechanics, anomalous transport is often studied through oscillator chains with local conservation laws and boundary reservoirs [7, 8, 9]. Our transport observable is the conserved action current of the resonant model, not the physical heat current of an anharmonic chain. Accordingly, any measured finite-size exponent must be interpreted as a numerical law for this model, not as a universal conductivity exponent.

This distinction also limits the local-equilibrium and fluctuation claims. Local equilibrium is a standard organizing principle in hydrodynamic descriptions of one-dimensional transport [9], but the Hamiltonian (1.3) contains a nonlocal mass term, so even equilibrium two-site

marginals are not strict Gibbs marginals of an isolated bond. We therefore test a weaker pair-marginal rescaling rather than claim local Gibbs convergence. Likewise, fluctuation relations for deterministic and stochastic nonequilibrium steady states [6, 10] motivate future entropy-production measurements, but finite-time action-current averages alone do not constitute such a test. A Gallavotti–Cohen analysis would require a separate boundary-energy accumulator and is outside the claims made here.

1.4 Contributions and organization

This paper is a numerical study of the NESS of the energy cascade model (1.5). After collecting the thermodynamic and spectral framework in sections 2.1 to 2.3, we present two groups of results.

The division between the two groups is deliberate. The long-chain experiments probe macroscopic statistical structure in a regime where no invariant-measure theory is currently available, whereas the short-chain computations provide a low-dimensional view of the stabilization and phase-locking mechanisms behind the rigorously understood three-mode NESS.

Section 3 treats the *long chain*, the regime left open by [2]. We characterize the nonequilibrium steady state through its action profile and an approximate local equilibrium in which adjacent-mode pair marginals match a rescaled equilibrium marginal at the local kinetic temperature, without asserting strict local-Gibbs structure. We also formulate the transport problem for the conserved action current, keeping it distinct from Hamiltonian-energy exchange with the baths.

Section 4 returns to the *three-mode chain* for a detailed study. We solve the stationary Fokker–Planck equation by a mesh-free, data-driven neural-network method [3], validating it against the exact Gibbs measure (1.7) in equilibrium; we then examine the nonequilibrium steady state, the phase-locking stabilization mechanism, and the observable-dependent slow relaxation visible in trajectory data. The numerical tools—a high-throughput SIMD Monte Carlo integrator, a kernel density estimator for the reference measure, and a trajectory-based backward solver for slow-mode diagnostics—are described as they arise.

2 Preliminary

This section collects the thermodynamic observables we measure for the long chain (section 2.1), the spectral (Fokker–Planck) framework for the short chain (section 2.2), and the numerical methods used to solve the stationary Fokker–Planck equation and to extract eigenfunctions of the generator (section 2.3), and the validation protocol for the long-chain Monte Carlo calculations (section 2.4).

2.1 Nonequilibrium thermodynamics of the chain

The Hamiltonian flow (1.4) conserves the total mass $M = \sum_k I_k$, and the evolution of each action obeys a discrete continuity equation

$$\dot{I}_j = J_j - J_{j+1}, \quad J_j := 4I_{j-1}I_j \sin \theta_j, \quad (2.1)$$

so that J_j is the *action current* flowing from mode $j - 1$ into mode j (with $J_1 = J_{n+1} = 0$ for the closed chain). In the nonequilibrium steady state (NESS) the expected actions are stationary, $\mathbb{E}[\dot{I}_j] = 0$, hence the mean current is uniform along the chain, $\mathbb{E}[J_j] \equiv \mathbb{E}[J(n)]$ for all interior j .

Definition 2.1 (Action conductivity and Fourier’s-law analogy). Assume $T_1 > T_n$, so the positive action-current direction points from the hot left bath toward the cold right bath. Let

$$\nabla T(n) := \frac{T_n - T_1}{n} < 0$$

denote the effective temperature gradient in that positive spatial direction. We say, by analogy with Fourier's law, that the chain has finite action conductivity κ if

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[J(n)]}{\nabla T(n)} = -\kappa \in (0, \infty). \quad (2.2)$$

Let $\kappa_n = -\mathbb{E}[J(n)]/\nabla T(n)$ be the finite-chain action conductivity. Anomalous transport corresponds to $\kappa_n \sim n^\alpha$ with $\alpha \neq 0$: $\alpha > 0$ is superdiffusive and $\alpha < 0$ (so $\kappa_n \rightarrow 0$) signals an insulator.

When one measures the actual Hamiltonian energy exchanged with the baths, two further observables characterize the NESS. The *entropy production rate* associated with the two heat baths is

$$\dot{S} = -\frac{\mathbb{E}[\dot{Q}_1]}{T_1} - \frac{\mathbb{E}[\dot{Q}_n]}{T_n} \geq 0, \quad (2.3)$$

where $\dot{Q}_1, \dot{Q}_n$ are the mean heat fluxes delivered by the baths; in the NESS the energy balance forces $\mathbb{E}[\dot{Q}_1] = -\mathbb{E}[\dot{Q}_n] =: \mathbb{E}[\dot{Q}]$ and $\dot{S} = \mathbb{E}[\dot{Q}](1/T_n - 1/T_1)$. The second law $\dot{S} \geq 0$ is refined by the steady-state *fluctuation theorem* of Gallavotti–Cohen [6]: under its usual chaoticity and large-deviation hypotheses, writing Σ_τ for the entropy produced over a window of length τ , the corresponding densities satisfy formally

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \ln \frac{p_\tau(a)}{p_\tau(-a)} = a. \quad (2.4)$$

In this paper the transport measurements below concern the conserved *action* current. A full bath-energy and entropy-production measurement is left as a separate numerical diagnostic and is not used as evidence for the main claims.

2.2 Fokker–Planck equation and spectral decomposition

Write $x = (\mathbf{I}, \boldsymbol{\varphi})$ and let the system (1.5) have drift $f(x)$ and diffusion matrix $\Sigma(x) = \sigma(x)\sigma(x)^\top$. The probability density $u(x, t)$ evolves by the (forward) Fokker–Planck, or Kolmogorov forward, equation $\partial_t u = \mathcal{L}^* u$, where

$$\mathcal{L}^* u = -\sum_i \partial_{x_i}(f_i u) + \frac{1}{2} \sum_i \partial_{x_i}^2 (\Sigma_{ii} u). \quad (2.5)$$

For our model the diffusion is diagonal and degenerate: only the bath modes carry noise, with

$$\frac{1}{2} \Sigma_{I_j I_j} = 4\gamma T_j I_j, \quad \frac{1}{2} \Sigma_{\varphi_j \varphi_j} = \gamma T_j / I_j, \quad j \in \{1, n\}, \quad (2.6)$$

and zero otherwise in the full $(\mathbf{I}, \boldsymbol{\varphi})$ coordinates. In reduced coordinates built from doubled bond angles, for instance $\theta_2 = 2(\varphi_2 - \varphi_1)$ at the left boundary, the angular diffusion coefficient is multiplied by 2^2 , giving $\frac{1}{2} \Sigma_{\theta_2 \theta_2} = 4\gamma T_1 / I_1$ (and similarly at the right boundary).

Remark 2.2 (Degeneracy / hypoellipticity). Because noise enters only at $j \in \{1, n\}$, Σ has rank 4 in the $2n$ -dimensional state space; the interior modes are forced only through the Hamiltonian coupling. Existence, uniqueness and ergodicity of the invariant measure for the $n = 3$ system are established in [2] via a Feynman–Kac–Lyapunov argument together with Hörmander's condition; the long chain $n > 3$ is left there as an open problem and is the subject of section 3.

The stationary density solves $\mathcal{L}^* u = 0$ with $\int u \, dx = 1$. The long-time relaxation toward it is governed by the (backward) generator $\mathcal{L}$, the formal adjoint of $\mathcal{L}^*$,

$$\mathcal{L} g = \sum_i f_i \partial_{x_i} g + \frac{1}{2} \sum_i \Sigma_{ii} \partial_{x_i}^2 g, \quad (\mathbb{E}_x[g(X_t)]) = (e^{t\mathcal{L}} g)(x). \quad (2.7)$$

Under ergodicity the spectrum of $\mathcal{L}$ lies in $\{\text{Re } \lambda \leq 0\}$ with a simple eigenvalue $\lambda_0 = 0$ (constant eigenfunction). Ordering the remaining eigenvalues by real part, the *spectral gap* $\lambda_R := \max\{\text{Re } \lambda : \lambda \neq 0\} < 0$ sets the slowest relaxation rate, and for an observable g orthogonal to constants

$$\mathbb{E}_x[g(X_t)] - \mathbb{E}_{\text{eq}}[g] \sim Q(x) e^{\lambda_R t} \quad (t \rightarrow \infty), \quad (2.8)$$

where Q is the corresponding (sub)dominant eigenfunction, $\mathcal{L}Q = \lambda_R Q$. A purely real λ_R indicates an overdamped relaxation; a nonzero imaginary part λ_I indicates damped oscillation with frequency λ_I .

2.3 Numerical methods

2.3.1 A data-driven neural-network Fokker–Planck solver

Classical grid-based PDE solvers are impractical for (2.5) in dimension $2n$: the domain is unbounded with no natural boundary condition, and the cost grows exponentially with dimension. We follow the mesh-free, data-driven solver of Zhai–Dobson–Li [3] (building on [4, 5]), which represents the stationary density by a neural network $\tilde{u}(x; \vartheta)$ trained against the unconstrained loss

$$\mathcal{J}(\vartheta) = \underbrace{\frac{1}{N_X} \sum_i (\mathcal{L}^* \tilde{u}(x_i; \vartheta))^2}_{\text{PDE residual } L_1} + \underbrace{\frac{1}{N_Y} \sum_j (\tilde{u}(y_j; \vartheta) - v(y_j))^2}_{\text{MC reference } L_2}, \quad (2.9)$$

where $\mathcal{L}^* \tilde{u}$ is evaluated by automatic differentiation and v is a low-accuracy Monte Carlo (MC) estimate of the density at reference points $\{y_j\}$. The operator term L_1 regularizes the network toward solutions of $\mathcal{L}^* u = 0$ and sharply reduces the amount and accuracy of MC data required; the method tolerates spatially uncorrelated noise in v [3].

We adapt the solver to (1.5) in two ways. First, the angles are periodic, so the network input is the *periodic encoding* $(\mathbf{I}, \cos \theta_1, \sin \theta_1, \dots)$ rather than the raw angles. Second, the diffusion (2.6) is degenerate (remark 2.2); we therefore include collocation points in the zero-density region and constrain them by the operator L_1 alone (no fabricated reference data there). The reference density v is obtained by a high-throughput MC simulation of (1.5) together with a kernel density estimate; see section 4.1.

2.3.2 Eigenfunctions of the generator

To probe (2.8) we estimate the relaxation rate and slow-mode amplitude visible to a chosen observable directly from trajectories. For a localized observable g (we use $g = \cos \theta_1$) we evolve a large ensemble of trajectories from each of many initial points x and record the conditional expectation $\mathbb{E}_x[g(X_t)]$ as a function of t . Fitting a late-time exponential window gives an observable-dependent estimate of the dominant decay rate visible to g ; if a damped oscillatory fit is also stable, it gives an estimate of an imaginary part. The fitted amplitude gives a pointwise sample $Q(x)$ of the associated slow mode. A second network is then trained to represent Q on the whole domain, using a data loss against the sampled $Q(x)$ together with a PDE loss enforcing the eigenvalue equation $\mathcal{L}Q = \lambda_R Q$. These fits are diagnostics rather than proof that the estimated rate is the window-independent spectral gap of the full generator. Results are reported in section 4.3.

2.4 Monte Carlo validation and uncertainty protocol

The long-chain histogram and reference-density calculations use the SIMD Euler–Maruyama workflow, while the burn-in and finite-time current studies use a separate double-precision canonical accumulator. The two data streams are analyzed separately and are not pooled. Current

uncertainties are computed across independent trajectory averages; equal subwindows of a trajectory are used only to diagnose finite-time drift, not as independent replicates.

Before a production dataset is used as evidence, the corresponding executable is checked for seeded reproducibility, agreement across thread counts, equal-temperature zero current, current reversal after swapping the baths, and sensitivity to timestep and burn-in. Scaling fits, when reported, must state the fitted chain-length window and attach uncertainty intervals obtained by resampling independent trajectories. These criteria define the acceptance protocol; quantitative outcomes are reported with the experiment to which they apply.

3 Statistical physics properties of the long cascade chain

The rigorous analysis of [2] treats the three-mode chain and lists the longer chains $n > 3$ as open. This section studies those long chains numerically under the Gibbs-preserving two-bath dynamics (1.6): the structure of the nonequilibrium steady state (section 3.1), the transport of the conserved action current and the associated finite-chain conductivity (section 3.2), and finite-time current fluctuations. We keep the terminology “action current” throughout, because (2.1) is a continuity equation for the conserved action $M = \sum_j I_j$; Hamiltonian energy exchange with the baths is a distinct observable.

Simulation. All long-chain data are produced by a single Euler–Maruyama integrator of (1.5) that processes $W = 16$ independent trajectories in lockstep (one SIMD register per arithmetic operation), with the phases wrapped to $[-\pi, \pi)$ at every step. Unless otherwise stated we use $\gamma = 0.1$ and the closed two-bath chain (1.6). The local-equilibrium histograms use long on-the-fly accumulation runs, while the current-scaling runs use an independent canonical current accumulator with double precision, deterministic trajectory seeds, four equal current blocks per trajectory, and explicit timestep and burn-in diagnostics.

3.1 Nonequilibrium steady state

Stabilization. We regard a run as stabilized when the sampled expectations (in particular the action profile $\langle I_j \rangle$ and the bond-angle statistics) cease to change with increasing sample size. Because the cascade transports energy slowly, the relaxation time is long and the burn-in must be chosen with care: a short run can report a spuriously uniform profile that has not yet developed the steady gradient. We return to this point below.

Action profile. Figure 1 shows the mean action profiles $\langle I_j \rangle$ for closed chains with $T_1 = 10$, $T_n = 2$ at $n = 25, 50, 100$. The profiles are monotone after a short boundary layer at the hot contact and show the sharpest drop at the cold bath. For $n = 25$, the profile decreases from $\langle I_1 \rangle = 0.738$ at the hot end to $\langle I_{13} \rangle = 0.573$ near mid-chain and $\langle I_{25} \rangle = 0.165$ at the cold contact. The corresponding endpoint values are $(0.514, 0.106)$ at $n = 50$ and $(0.362, 0.073)$ at $n = 100$. These profile estimates are the data used below to define the local kinetic temperature in the LTE comparison.

Cascade geometry. The interpretation of the chain as an energy cascade rests on the frequency embedding of [1]: each generation of the resonant “nuclear family” construction places a group of modes on $\mathbb{Z}^2$ by exact rational (Pythagorean) rotations, so that the abstract chain position j corresponds to a concrete set of frequencies Λ_j . Figure 2 plots the locations of the first five generations used in the CKSTT construction. The open-chain terminal-energy scaling is not used as evidence in the present paper; establishing it would require a separate convergence study under the same corrected thermostat used for the closed-chain transport measurements.

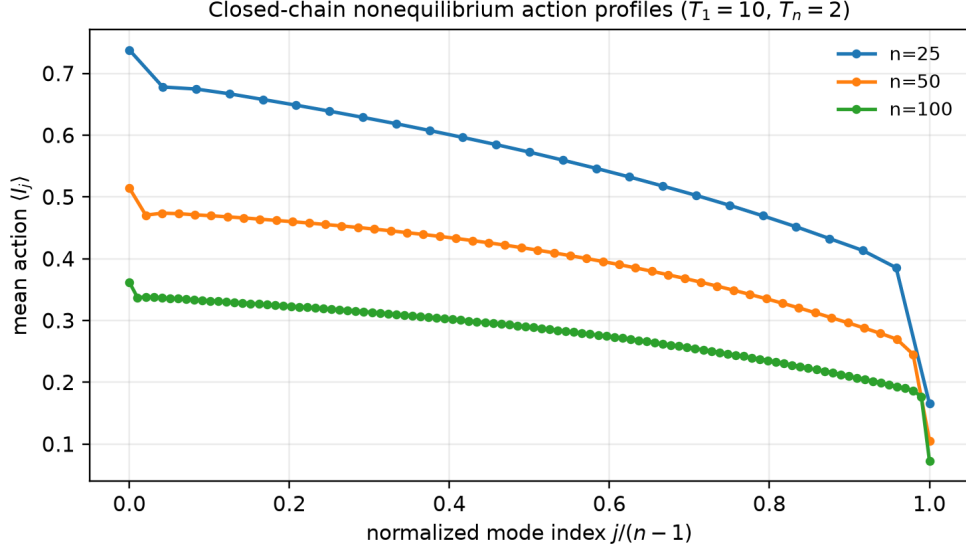


Figure 1: Steady-state action profiles of the long closed chain. The horizontal coordinate is normalized as $j/(n-1)$ to compare $n = 25, 50, 100$ on the same axis. The sample counts are 5.07×10^8 , 1.01×10^9 , and 4.06×10^9 , respectively.

Local thermodynamic equilibrium. A central question is whether, despite the global nonequilibrium drive, adjacent modes in the bulk are described *locally* by an equilibrium measure. Let $q(I_a, I_{a+1}, \theta_a)$ denote the joint pair density in the NESS, with $(I_a, I_b) = (I_j, I_{j+1})$, and let $p_T(I_a, I_{a+1}, \theta_a)$ be the corresponding marginal of the equilibrium measure at temperature T , whose full density is proportional to $\exp[-H/(2T)]$ under the convention of remark 1.1 (integrating out all other modes). We accumulate q and p_T on 80^3 histograms over $[0, 4]^2 \times [-\pi, \pi]$ from $\sim 10^9$ samples per site, at interior pairs of the closed chain ($T_1 = 10, T_n = 2$) for $n = 25, 50, 100$.

First, neither q nor p_T equals the two-mode Gibbs density $\exp[-(\frac{1}{2}(I_a + I_b)^2 - \frac{1}{4}(I_a^2 + I_b^2) + I_a I_b \cos \theta)]$: comparing $-\log(\text{PDF})$ to this form, the NESS and the equilibrium marginal deviate in the *same* way (both rise along the axes where the two-mode form is flat, and their θ -modulation is weaker). Since the equilibrium marginal already deviates identically, the discrepancy comes from marginalizing the nonlocal term $\frac{1}{2}(\sum_j I_j)^2$, not from the nonequilibrium drive: there is no strict local Gibbs structure, even at equilibrium.

Second, the NESS marginal is nonetheless well approximated by a *rescaled* equilibrium marginal. We fit $\log q = x \log p_6 + c$ by weighted least squares, weighting each bin by its count (the inverse variance of the log-density for multinomial counts); statistical errors on x are below 3×10^{-4} and are negligible against the systematic spread discussed below. Restricted to the core ($I_a, I_b < 2.5$) the fit is excellent, with weighted R^2 from 0.986 to 0.999 (table 1), and the rescaled p_6^x matches q closely (fig. 3).

We caution against reading the slope as a temperature through the naive relation $6/x$. A control fit between two simulated equilibria, $\log p_{7.12} = x \log p_6 + c$, returns $x = 0.926$ rather than the naive $6/7.12 = 0.843$: the marginals do not scale as $p_T = p_6^{6/T}$, because the mean field $\langle \sum_j I_j \rangle \propto \sqrt{nT}$ shifts with temperature. Hence $6/x$ systematically underestimates the effective temperature.

Third, the effective temperature is the local kinetic temperature. Define

$$T_{\text{kin}}(a) = 6 \frac{\langle I_a \rangle_{\text{NESS}}}{\langle I_a \rangle_{\text{eq}}}$$

directly from the action profiles, independent of the marginal fit. Matching q to an actual simulated equilibrium confirms this scale: at $a/n = 0.48$ the NESS marginal fitted against the

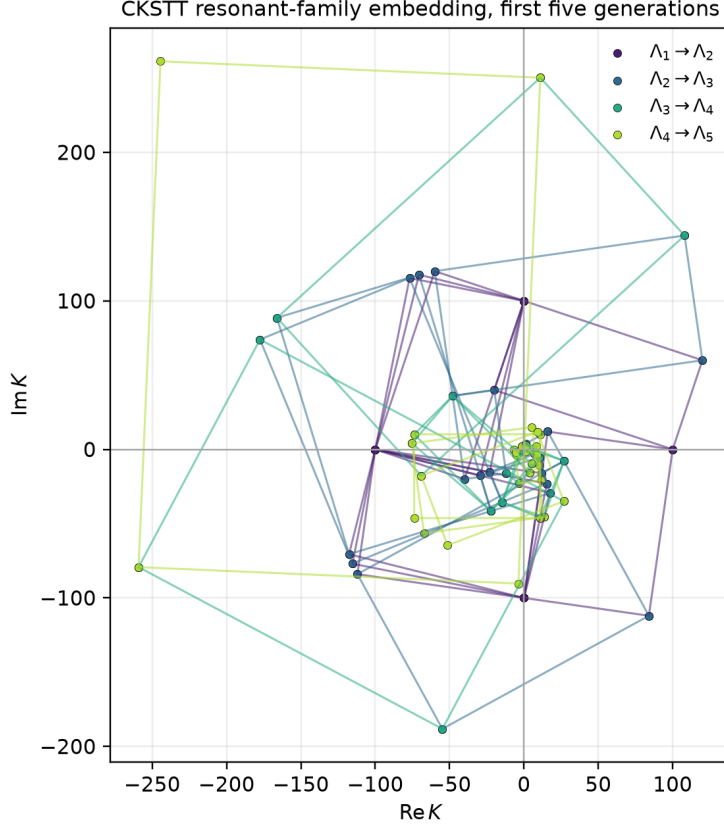


Figure 2: Cascade geometry: the first five generations in the CKSTT resonant frequency embedding. Points are Fourier modes $K \in \mathbb{Z}^2$ (or exact rational intermediate coordinates in the displayed construction), and segments indicate resonant family transitions between adjacent generations.

true $p_{7.12}$ gives $x = 1.01$, i.e. $q \approx p_{7.12}$, within a few percent of $T_{\text{kin}} = 6.85$ at that site (and consistent to 0.2% with the q - p_6 and $p_{7.12}$ - p_6 fits combined). The kinetic temperature, the equilibrium-matching temperature, and the naive $6/x$ agree to within 5–10%, the spread reflecting the residual departure of q from a true equilibrium marginal. To leading order, then, the dependence of the effective temperature on $(T_L, T_R, a/n)$ is simply the shape of the temperature profile; across boundary pairs of equal mean $((8, 4), (9, 3), (10, 2))$ it varies monotonically with the imposed gradient.

Fourth, the approximation improves as the chain lengthens. The control fit between two equilibria ($p_{7.12}$ vs. p_6) gives a weighted $R^2 = 0.9997$, the ceiling attainable within the equilibrium family. At fixed a/n the core R^2 rises toward this ceiling with n —most clearly at the cold, large-gradient sites ($a/n = 0.72$: $0.986 \rightarrow 0.992 \rightarrow 0.998$ for $n = 25, 50, 100$) and at mid-chain ($a/n = 0.48$: $0.994 \rightarrow 0.999 \rightarrow 0.999$), while the hot site ($a/n = 0.24$) is already near the ceiling at $n = 25$. A small gap persists at all n (about 0.001–0.002 at $n = 100$); in this sense $q \rightarrow p_{\beta_0}$ in the long-chain interior without becoming exactly equal to it.

The residual surfaces provide a direct diagnostic of where a single-slope rescaling is least accurate. For this visualization, `compare_residual.m` refits $\log q = c + x \log p_6$ using bins for which both histograms contain more than 50 counts, then displays the residual over the two action coordinates. This plotting fit is distinct from the core-weighted fit in table 1, and its slope is not used as a second effective-temperature estimator. Figure 3 shows the resulting three-dimensional meshes for $n = 15, 25, 50$.

a/n	n	x	R_w^2	$6/x$	T_{kin}
0.24	25	0.8203	0.9970	7.31	7.68
	50	0.8269	0.9984	7.26	7.79
	100	0.8429	0.9967	7.12	7.82
0.48	25	0.9360	0.9943	6.41	6.85
	50	0.9308	0.9988	6.45	7.03
	100	0.9304	0.9989	6.45	7.01
0.72	25	1.0933	0.9862	5.49	5.80
	50	1.0610	0.9919	5.65	6.04
	100	1.0412	0.9980	5.76	6.02
<i>Controls</i> ($n = 25$):					
$p_{7.12}$ vs. p_6 (ceiling)		0.9261	0.9997	—	
NESS vs. $p_{7.12}$ ($a/n=0.48$)		1.0093	0.9944	—	

Table 1: Weighted least-squares fit $\log q = x \log p_6 + c$ over the core ($I_a, I_b < 2.5$): slope x , weighted R^2 , the naive temperature $6/x$, and the independently measured kinetic temperature T_{kin} , at fixed a/n for $n = 25, 50, 100$. Statistical errors on x are below 3×10^{-4} , negligible against the 5–10% spread among temperature estimators. The two control fits (two equilibria, giving the ceiling; and NESS against the matched equilibrium $p_{7.12}$, $x \approx 1$) are listed below. The naive $6/x$ lies systematically below T_{kin} because the marginals do not follow the naive Gibbs scaling.

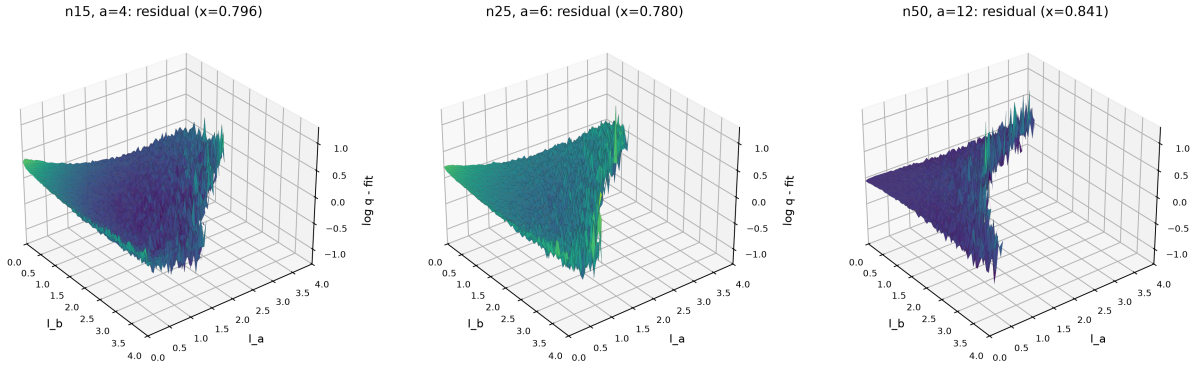


Figure 3: Three-dimensional LTE residual meshes for $n = 15, 25, 50$. The plotted quantity is $\log q - (c + x \log p_6)$ over the two action coordinates, using the count-thresholded plotting fit described in the text; its slope x is shown above each panel.

3.2 Action-current conductivity

TODO

3.3 Bath-energy entropy production: scope of the present numerics

The action-current accumulator does not record the stochastic Hamiltonian energy exchanged with each reservoir. A numerical evaluation of (2.3) therefore requires a separate boundary-energy accumulator for $\dot{Q}_1$ and $\dot{Q}_n$. No entropy-production or Gallavotti–Cohen result is claimed from the present action-current data.

4 Detailed property of the short chain

We now turn to the smallest nontrivial cascade, the three-mode system $n = 3$, for which existence, uniqueness and ergodicity of the NESS are rigorously known [2]. The Hamiltonian and

bath of section 3 depend on the phases only through $\theta_1 = 2(\varphi_1 - \varphi_2)$ and $\theta_3 = 2(\varphi_3 - \varphi_2)$, so the dynamics reduces from the six variables $(\mathbf{I}, \boldsymbol{\varphi})$ to the five-dimensional state $x = (I_1, I_2, I_3, \theta_1, \theta_3)$, in which the middle mode I_2 carries no bath. This low dimension lets us solve the stationary Fokker–Planck equation directly and interrogate the invariant measure (section 4.1), the mechanism that stabilizes the NESS (section 4.2), and the spectrum of the generator (section 4.3).

4.1 Numerical solution of the nonequilibrium steady state

In the reduced coordinates the stationary density solves $\mathcal{L}^*u = 0$, with $\mathcal{L}^*$ as in (2.5): the drift is the $n = 3$ specialization of (1.5), and the (degenerate) diffusion (2.6) acts only on (I_1, θ_1) and (I_3, θ_3) with coefficients $4\gamma T_1 I_1$, $4\gamma T_1/I_1$ and $4\gamma T_3 I_3$, $4\gamma T_3/I_3$. We solve it with the neural-network method of section 2.3. The reference density v is produced by a long SIMD Monte Carlo integration of (1.5) followed by a kernel density estimate on a grid of boxes in x ; boxes in which the Monte Carlo records no sample are retained as zero-density collocation points and are constrained by the operator term L_1 alone. The network outputs the log-density, $u(x)$ with $\rho = \exp(\sigma_0 u + \mu_0)$ for fixed normalization constants (μ_0, σ_0) , takes the periodic encoding $(I_1, I_2, I_3, \cos \theta_1, \sin \theta_1, \cos \theta_3, \sin \theta_3)$ as input, and is trained in two phases: a data-only pre-training against $\{v(y_j)\}$, followed by joint training in which the weight of the PDE residual L_1 is ramped on gradually.

Validation against the equilibrium Gibbs measure. When $T_1 = T_3 = T$ the exact invariant density is the Gibbs measure (1.7), which provides a stringent test. Training at $T = 5$ and comparing the network density to (1.7) on the slices $I_1 = I_2 = I_3 = I$ (fig. 4), the mean relative error is 1.5%, 2.6% and 6.4% at $I = 0.5, 1, 2$ respectively, with the pointwise log-ratio $\log(\rho_{\text{NN}}/\rho_{\text{Gibbs}})$ confined to a narrow band. The solver thus recovers the known equilibrium measure across the range of actions that carry appreciable probability.

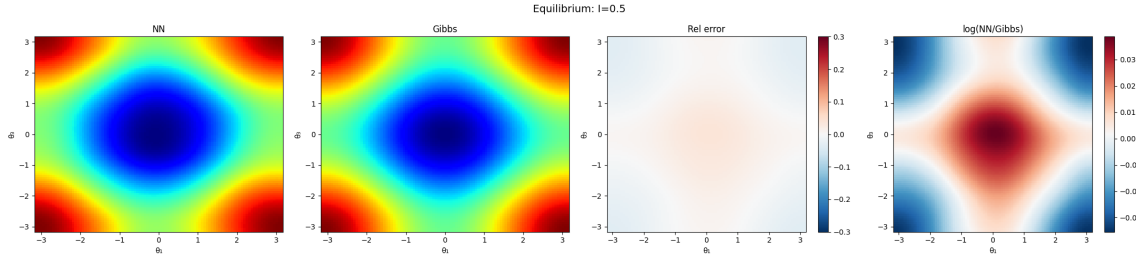


Figure 4: Equilibrium validation of the neural-network Fokker–Planck solver. At $T_1 = T_3 = 5$ and $I_1 = I_2 = I_3 = 0.5$, the network density $\rho(\theta_1, \theta_3)$ (left) is compared with the analytic Gibbs measure (centre-left); the relative error (centre-right) and log-ratio (right) stay within a few percent across the cell.

Nonequilibrium steady state. With distinct bath temperatures $T_1 = 2$, $T_3 = 8$ the invariant measure is no longer of Gibbs form. Figure 5 shows the network density on the angular plane (θ_1, θ_3) at several values of the common action: the distribution concentrates around a phase-locked configuration and is visibly distorted relative to the equilibrium measure. The departure from equilibrium is quantified in section 4.2.

4.2 Mechanism of stabilization

The NESS is stabilized by a phase-locking mechanism that operates differently in the high- and low-energy regimes [2]. At high energy the bond angles are driven to a stable equilibrium near $2\pi/3$, escaping the unstable configuration at $4\pi/3$; at low energy the angle θ_3 remains near

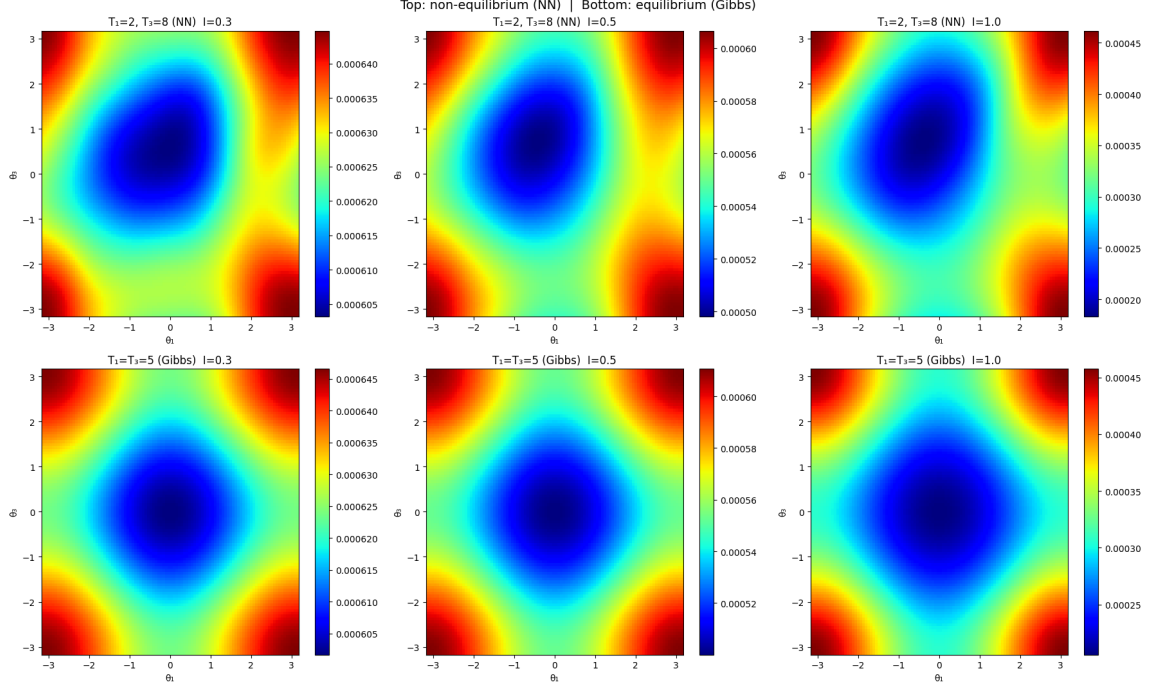


Figure 5: Nonequilibrium steady-state angular density of the three-mode chain. Top row: network density $\rho(\theta_1, \theta_3)$ under the gradient $T_1 = 2, T_3 = 8$ at $I = 0.3, 0.5, 1.0$; bottom row: the equilibrium ($T_1 = T_3 = 5$) density at the same I . The nonequilibrium density is tilted and shifted off-centre, breaking the $\theta_1 \leftrightarrow \theta_3$ symmetry of the equilibrium case.

$4\pi/3$, which prevents the middle action I_2 from collapsing to zero and keeps the cascade alive. The reduced four-dimensional subsystem $(I_1, I_3, \theta_1, \theta_3)$, obtained in the low-energy limit and equipped with the $\gamma(T - I^3)$ thermostat of [2], isolates this mechanism; we solve its stationary Fokker–Planck equation by the same network method and recover the predicted concentration of θ_3 .

We retain two source-traced diagnostics of the nonequilibrium drive at $T_1 = 2, T_3 = 8$: symmetry breaking in the angular density and the phase-locking branch predicted by the reduced equation.

(i) *Symmetry breaking.* At equilibrium the density is symmetric under $\theta_1 \leftrightarrow \theta_3$. Out of equilibrium this symmetry is broken: the relative asymmetry $|\rho(\theta_1, \theta_3) - \rho(\theta_3, \theta_1)|/(\rho(\theta_1, \theta_3) + \rho(\theta_3, \theta_1))$ is visibly nonzero on the high-density part of the angular plane (fig. 6). We report the qualitative effect here and reserve the precise max/mean asymmetry for the source-traced grid recomputation, since it depends sensitively on the density mask and normalization.

(ii) *Phase-locking angle.* The most probable bond angle is shifted from its equilibrium value to

$$\theta_0 = \pi - \arcsin \frac{\sqrt{4\gamma^2 + 3} - \gamma}{2(\gamma^2 + 1)}, \quad (4.1)$$

the stable branch that tends to $2\pi/3$ as $\gamma \rightarrow 0$. We use the density plots qualitatively to display concentration near the stable sector; no quantitative peak-location table is used as an independent result here.

Remark 4.1. Equivalently, writing $s = \sin \theta_0$, the reduced angular fixed-point condition gives the quadratic

$$4(1 + \gamma^2)s^2 + 4\gamma s - 3 = 0.$$

The positive root is

$$s = \frac{\sqrt{4\gamma^2 + 3} - \gamma}{2(1 + \gamma^2)}.$$

The inverse sine has two branches. The branch in (4.1) lies in $(\pi/2, \pi)$ and tends to $2\pi/3$ as $\gamma \rightarrow 0$, matching the stable high-energy phase-locking equilibrium; the complementary branch tends to $\pi/3$ and is not the observed stable lock.

Finally, current conservation through the unforced middle mode gives the balance condition $\mathbb{E}[I_2 I_1 \sin \theta_1] + \mathbb{E}[I_2 I_3 \sin \theta_3] \approx 0$ in a stationary state. We use this identity as a rerun diagnostic for the solver, but do not report it as a separate quantitative result here.

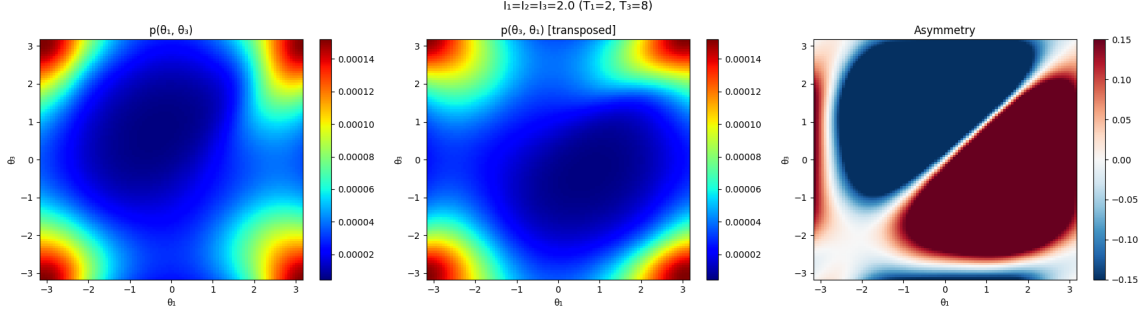


Figure 6: $\theta_1 \leftrightarrow \theta_3$ symmetry breaking in the nonequilibrium steady state. At $I_1 = I_2 = I_3 = 2$ and $T_1 = 2, T_3 = 8$, the network density $\rho(\theta_1, \theta_3)$ (left) and its transpose (centre) differ; the relative asymmetry $(\rho - \rho^T)/(\rho + \rho^T)$ (right) is shown with the colour scale clipped at ± 0.15 for readability. Since unmasked max/mean asymmetry is sensitive to low-density regions and network normalization, this panel is used as qualitative evidence of symmetry breaking rather than as a standalone quantitative estimator.

4.3 Eigenfunction of the generator

To probe the relaxation spectrum we apply the trajectory method of section 2.3 with the localized observable $g = \cos \theta_1$. From each of 4096 initial points we evolve an ensemble and record $\mathbb{E}_x[\cos \theta_1(X_t)]$; fitting the decay (2.8) returns, at each initial point, an estimate of the observable-dependent relaxation rate and amplitude.

At $\gamma = 0.1$, pure-exponential fits

$$\mathbb{E}_x[\cos \theta_1(X_t)] \approx A(x)e^{\lambda t} + c(x)$$

over the full recorded window $t \in [0, 5]$ retain 2895 of 4096 curves under the filter $-5 < \lambda < 0$, with median

$$\lambda_{\text{diag}} \approx -0.934. \quad (4.2)$$

This number is reproducible from the saved backward-Monte-Carlo curves, but it is not a window-independent spectral gap estimate: the retained median shifts from -1.60 on $t \in [0, 2]$ to -0.68 on $t \in [0.5, 5]$ (fig. 7). We therefore use λ_{diag} as a diagnostic relaxation rate for the observable $\cos \theta_1$, not as a resolved statement about the full generator spectrum.

The fitted amplitudes provide pointwise samples $Q(x)$ of the corresponding slow-mode surrogate. A second network is trained to represent Q subject to the eigenvalue equation $\mathcal{L}Q = \lambda_{\text{diag}}Q$, using the generator $\mathcal{L}$ of (2.7) (without the divergence terms that distinguish $\mathcal{L}^*$). The reconstruction is only moderately accurate: the data root-mean-square error is 0.38 (relative error 0.71) and the median normalized PDE residual $|\mathcal{L}Q - \lambda_{\text{diag}}Q|/|Q|$ is 0.35. We report these candidly: the fitted amplitude data contain both signs and are reconstructed from intrinsically

noisy backward Monte Carlo data, making this the hardest object computed in this work; the agreement is nonetheless sufficient to exhibit the qualitative structure of Q on representative slices (fig. 8).

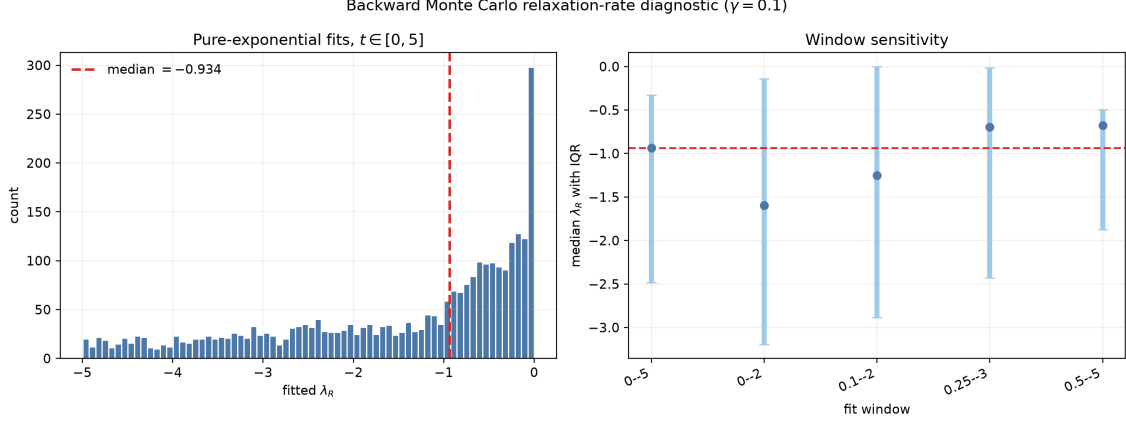


Figure 7: Backward-Monte-Carlo relaxation-rate diagnostic at $\gamma = 0.1$, using the observable $\cos \theta_1 = \cos 2(\varphi_1 - \varphi_2)$. Left: retained pure-exponential rates from 4096 initial conditions over $t \in [0, 5]$; 2895 fits satisfy $-5 < \lambda < 0$, with median $\lambda_{\text{diag}} \approx -0.934$. Right: median and interquartile range across several fitting windows, showing the window sensitivity that motivates treating this quantity as a diagnostic rate rather than as a high-accuracy spectral gap.

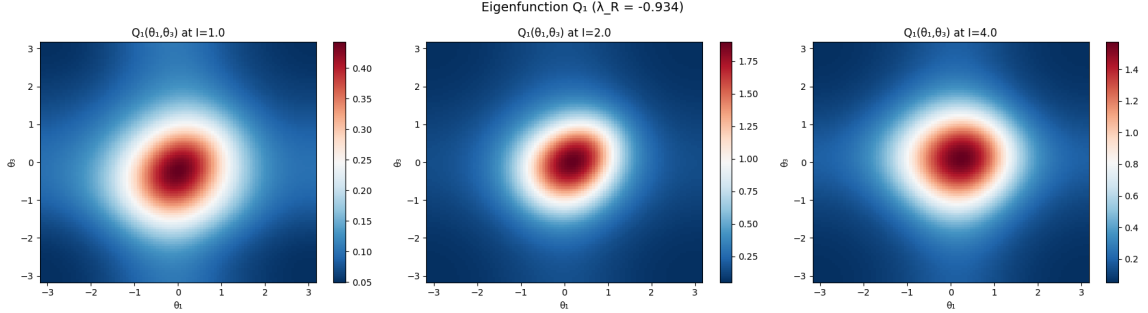


Figure 8: Generator slow-mode surrogate. Network reconstruction of $Q(\theta_1, \theta_3)$ at $I_1 = I_2 = I_3 = 1, 2, 4$ (fitted $\lambda_{\text{diag}} \approx -0.934$). The fitted amplitude data contain both signs; the displayed slices show the positive lobe concentrated near $\theta_1 = \theta_3 = 0$. As noted in the text, the surrogate is reconstructed from noisy backward Monte Carlo data and is shown to exhibit the qualitative structure of Q rather than as a high-accuracy spectral computation.

5 Conclusion and limitations

TODO

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